Tiny Language Model Based Context-Aware Resource Allocation for 6G Semantic Communication Systems

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***Abstract*—This paper studies semantic-aware resource allocation in 6G systems where each user message is assigned an importance score by a lightweight on-device TinyML model. The objective is to maximise semantic utility while ensuring queue stability under stochastic arrivals and fading channels. Fixing semantic scores per time-slot renders the problem jointly concave, enabling a closed-form Semantic Water-Filling solution with $\mathcal{O}(N \log N)$ complexity suitable for real-time operation. A Lyapunov Drift-Plus-Penalty controller guarantees an $\mathcal{O}(1/V)$ utility gap and $\mathcal{O}(V)$ queue bound.**

Simulation results show that in normal traffic regimes, the proposed method matches baseline schedulers in utility, while under heterogeneous load it significantly reduces delay and stabilises queue variance. This establishes the primary contribution as robustness under load imbalance rather than universal superiority. Additional extensions (ADMM, Nash equilibrium, SCA, SDP) are included with explicit validity domains. A hardware-aware TinyML compression pipeline reduces model energy by 19$\times$ with negligible accuracy loss.

***Index Terms*—6G, semantic communication, TinyML, convex optimisation, Lyapunov drift-plus-penalty, water-filling, ADMM, Nash equilibrium, successive convex approximation, edge AI, quantisation-aware training.**

I. Introduction

Sixth-generation (6G) wireless networks are expected to support a new communication paradigm in which the meaning of a transmitted message, rather than mere bit accuracy, governs resource allocation. In semantic communication, each message M_i is associated with an importance score $S_i \in [0, 1]$ produced by a lightweight neural model deployed at the network edge or on the user device. Allocating radio resources proportionally to S_i improves the downstream utility of the received information while respecting stringent delay and energy constraints.

The joint optimisation of power and bandwidth under semantic weights, queue dynamics, and stochastic fading is generally non-convex and non-stationary. This paper provides:

1) A minimal, self-contained solution for a single centralised base station (BS) based on Lyapunov Drift-Plus-Penalty (DPP) control and closed-form Semantic Water-Filling.
2) Principled extensions for distributed BSs (ADMM), selfish devices (Nash Equilibrium), power-coupled scores (SCA), and MIMO beamforming (SDP), each with explicit trigger conditions and validity domains.
3) A TinyML compression pipeline that decouples model inference from online optimisation.

Contribution framing. Unlike conventional works that focus on average throughput gains, this paper targets robustness under non-ideal traffic conditions. In homogeneous traffic regimes, all schedulers achieve comparable utility due to symmetric channel statistics. The proposed method differentiates itself under heterogeneous load, where queue-aware control significantly reduces delay and stabilises backlog variance. Therefore, the contribution is not universal improvement, but consistent performance under stress conditions.

The rest of the paper is organised as follows. Section II states the system assumptions. Section III formulates the joint optimisation problem. Section IV presents the core three-step solution. Sections V–XI detail each method and its validity domain. Section XIV reports simulation results, and Section XV concludes.

II. System Assumptions

All theoretical claims in this paper are valid only within the domains stated in Table I. The most critical assumption—that the semantic score S_i is fixed as an exogenous input at the start of each time-slot—is what renders the core problem

convex. When this assumption is violated (i.e., S_i depends on P_i), Method C (SCA) must replace the standard solver.

III. Problem Formulation

A. Joint Semantic-CSI Resource Allocation (P0)

Let $\mathcal{N} = \{1, \ldots, N\}$ be the set of users, $P_i(t)$ the transmit power, $B_i(t)$ the allocated bandwidth, $h_i(t)$ the complex channel coefficient, and N_0 the noise power spectral density. The Shannon rate for user i is

$$R_i(t) = B_i(t) \log_2\left(1 + \frac{P_i(t)|h_i(t)|^2}{N_0 B_i(t)}\right). \tag{1}$$

The semantic score is $S_i = \sigma(f_\theta(M_i)) \in [0, 1]$, where f_θ is the TinyML model and σ is the sigmoid activation.

The joint optimisation problem is:

$$\textbf{(P0)} \quad \underset{P_i, B_i}{\text{maximise}} \sum_{i \in \mathcal{N}} S_i(\theta, M_i) \cdot R_i(t) \tag{2}$$

$$\text{(C1)} \quad \sum_i P_i \le P_{\max}, \quad P_i \ge 0 \tag{3}$$

$$\text{(C2)} \quad \sum_i B_i \le B_{\text{total}}, \quad B_i > 0 \tag{4}$$

$$\text{(C3)} \quad Q_i(t{+}1) = \big[Q_i(t) - \mu_i(t)\big]^+ + \lambda_i(t) \tag{5}$$

$$\text{(C4)} \quad \mathbb{E}[Q_i(t)] < \infty \quad \forall i \tag{6}$$

$$\text{(C5)} \quad S_i = \sigma(f_\theta(M_i)) \in [0, 1], \text{ fixed per slot.} \tag{7}$$

B. Sources of Difficulty

Problem (P0) is hard to solve directly for four reasons:

1) **Bilinear coupling.** $S_i \times R_i$ is neither linear nor jointly convex. Fix: treat S_i as an input each slot $\to$ convex.
2) **Queue floor operator** $[\cdot]^+$. Breaks smooth differentiation; handled by the Lyapunov method.
3) **Random channel** $h_i(t)$. Decisions made without future knowledge; Lyapunov method requires no forecasting.
4) **Binary scheduling** (if used). $x_i \in \{0, 1\}$ makes it combinatorial; LP relaxation + randomised rounding applies.

IV. Core Three-Step Solution

This three-step approach is sufficient for a centralised single-BS deployment. All subsequent methods are triggered only by specific deployment scenarios.

Step 1 — Compute Semantic Scores. A TinyML model processes message M_i offline or from cache: $S_i = \sigma(f_\theta(M_i)) \in [0, 1]$. Once S_i is fixed, the objective (2) becomes jointly concave in (P, B), enabling efficient convex solvers.

Step 2 — Lyapunov DPP Control. Each time-slot, solve:

$$\underset{P_i, B_i}{\text{minimise}} \; -V \sum_i S_i R_i + \sum_i Q_i(\lambda_i - \mu_i) \quad \text{s.t.} \; \sum_i P_i \le P_{\max}. \tag{8}$$

Guarantee: $\mathbb{E}[\text{utility}] \ge \text{optimal} - B/V$ and $\mathbb{E}[Q_i] = \mathcal{O}(V)$.

Step 3 — Semantic Water-Filling. Applying KKT conditions to (8):

$$P_i^* = \left[\frac{S_i}{\mu} - \frac{N_0 B_i}{|h_i|^2}\right]^+, \tag{9}$$

Algorithm 1 Lyapunov DPP Resource Allocation (Per Slot)

Require: $h_i(t)$, $\lambda_i(t)$, $\mathbf{Q}(t)$, S_i (cached), V, $P_{\max}$

Observe channel $h_i(t)$ and arrivals $\lambda_i(t)$
Retrieve S_i from TinyML model (offline / cached)
Solve: $\min -V \sum_i S_i R_i + \sum_i Q_i(\lambda_i - \mu_i)$; s.t. $\sum_i P_i \le P_{\max}$ *(closed-form via Eq. (9))*
Transmit with optimal $\{P_i^*, B_i^*\}$
Update: $Q_i(t{+}1) = [Q_i(t) - \mu_i]^+ + \lambda_i(t)$

where $\mu = 1/\lambda$ is the water level found by bisection in $\mathcal{O}(\log(1/\varepsilon))$ steps. Total complexity: $\mathcal{O}(N \log N)$—no solver required in real-time deployment.

Table II maps each deployment scenario to the required methods.

V. Lyapunov Drift-Plus-Penalty Control

A. Formulation

Define the Lyapunov function measuring total queue backlog energy:

$$L(\mathbf{Q}) = \tfrac{1}{2} \sum_i Q_i^2(t). \tag{10}$$

The one-slot conditional drift is $\Delta(t) = \mathbb{E}[L(\mathbf{Q}(t{+}1)) - L(\mathbf{Q}(t)) \mid \mathbf{Q}(t)]$, which is bounded above as:

$$\Delta(t) \le B + \sum_i Q_i \cdot \mathbb{E}[\lambda_i - \mu_i \mid \mathbf{Q}(t)]. \tag{11}$$

The DPP objective penalises the drift while maximising utility:

$$\underset{P_i, B_i}{\text{minimise}} \; \Delta(t) - V \sum_i S_i(t) R_i(t), \tag{12}$$

where $V > 0$ is a control parameter trading utility against queue length.

B. Per-Slot Algorithm

C. Theorem (Neely 2010)

$$\mathbb{E}[\text{utility}] \ge \text{optimal} - \frac{B}{V} \quad \text{and} \quad \mathbb{E}[Q_i] = \mathcal{O}(V). \tag{13}$$

The V trade-off is unavoidable: higher V improves utility but increases the average queue. Choose V to match your delay budget.

D. Convexity After Fixing S_i

The CVXPY formulation below validates the convex structure offline.

Listing 1. CVXPY DCP Formulation (offline validation)

```
import cvxpy as cp, numpy as np

P = cp.Variable(N, nonneg=True)
# power per user
B = cp.Variable(N, nonneg=True)
# bandwidth per user

# SNR per user (perspective of log1p)
```

TABLE I
SYSTEM ASSUMPTIONS AND VIOLATION CONSEQUENCES

Applies To	Assumption Required	What Breaks When Violated
All Methods	Channel $h_i(t)$ follows a stationary (ergodic) distribution	Lyapunov drift bounds require stationarity; breaks under adversarial channels
All Methods	Arrival process $\lambda_i(t)$ is i.i.d. or ergodic	Queue-stability guarantee breaks under non-stationary traffic
Convex Solver	S_i is fixed at the start of each slot (treated as input, not variable)	If S_i depends on P_i, the problem is non-convex; use SCA (Method C)
ADMM	Backhaul links are reliable and low-latency (< 1 ms)	Delayed or out-of-order updates break convergence; speed depends on ρ and latency
Nash Eq.	Utility $u_i = S_i R_i - c_i P_i$ is concave in P_i; diagonal strict concavity holds (Rosen 1965); interference fixed during each user's update	Best-response dynamics fail to converge with non-convex coupling
SCA	Objective has Lipschitz-smooth gradient; surrogate satisfies conditions A1–A4	Poor starting point or non-smooth objectives can cause divergence

TABLE II
DEPLOYMENT SCENARIO TO REQUIRED METHODS MAPPING

Scenario	Required	Skip
Single centralised BS	Lyapunov + WF	ADMM, Nash, SDP
Distributed multi-BS	Lyapunov + ADMM	SDP
Selfish IoT devices	Lyapunov + Nash	SDP
Dynamic $S_i = f(P_i)$	Lyapunov + SCA	SDP
MIMO beamforming	SDP relaxation	Nash, ADMM
Edge device (low memory)	HAPE + BitNet QAT	SDP

```
SNR = cp.multiply(P, h) * cp.inv_pos(
        cp.multiply(N0, B))

# Shannon capacity
R = cp.multiply(B, cp.log1p(SNR) / np.log(2))

# S_scores fixed this slot (exogenous input)
obj  = cp.Maximize(S_scores @ R)
cons = [cp.sum(P) <= P_max,
        cp.sum(B) <= B_total]

prob = cp.Problem(obj, cons)
prob.solve(solver=cp.ECOS)
# Complexity: O(N^{1.5} eps^{-1}) -- interior-point
```

Table III summarises the convexity status of each term.

VI. SEMANTIC WATER-FILLING

Setting $\partial\mathcal{L}/\partial P_i = 0$:

$$\frac{S_i|h_i|^2}{N_0 B_i + P_i|h_i|^2} - \lambda = 0 \;\Rightarrow\; P_i^* = \left[\frac{S_i}{\mu} - \frac{N_0 B_i}{|h_i|^2}\right]^+, \quad (14)$$

where $\mu = 1/\lambda$ is the water level found by bisection. Users with higher S_i receive a higher effective water level and hence more power.

TABLE III
CONVEXITY STATUS OF PROBLEM TERMS

Term	Status	Resolution
Shannon Rate $R_i(P_i, B_i)$	Concave ✓	Exploitable by solver
Power budget $\sum P_i \le P_{\max}$	Convex ✓	Linear constraint
Queue drift $\frac{1}{2}\sum Q_i^2$	Convex ✓	Quadratic Lyapunov
$S_i = \sigma(WM_i + b)$	Non-convex ×	Use SCA if S_i dynamic
Product $S_i \times R_i$	Non-convex ×	Resolved by fixing S_i

Classical water-filling (no semantics) allocates power based on channel quality only: $P_i^* = [\mu - N_0/|h_i|^2]^+$. Semantic Water-Filling (14) extends this by weighting each user's level by S_i, preserving the $\mathcal{O}(N \log N)$ complexity while prioritising semantically important messages.

VII. METHOD C: SUCCESSIVE CONVEX APPROXIMATION (SCA)

Trigger condition: $S_i = f(P_i)$—semantic score depends on power level.

A. Surrogate Conditions

The surrogate function $\tilde{f}(\mathbf{P}; \mathbf{P}^k)$ must satisfy:

$$\text{(A1) } \tilde{f}(\mathbf{P}^k; \mathbf{P}^k) = f(\mathbf{P}^k) \quad (15)$$

$$\text{(A2) } \nabla\tilde{f}(\mathbf{P}^k; \mathbf{P}^k) = \nabla f(\mathbf{P}^k) \quad (16)$$

$$\text{(A3) } \tilde{f}(\mathbf{P}; \mathbf{P}^k) \text{ is a concave lower bound of } f \quad (17)$$

$$\text{(A4) Surrogate has Lipschitz-smooth gradient.} \quad (18)$$

B. Update Rule

$$\tilde{R}(\mathbf{P}) = R(\mathbf{P}^k) + \nabla R(\mathbf{P}^k)^\top(\mathbf{P} - \mathbf{P}^k) \quad (19)$$

$$\mathbf{P}^{k+1} = \alpha\mathbf{P}^k + (1-\alpha)\mathbf{P}^* \quad (20)$$

$$\text{Stop when: } \|\mathbf{P}^{k+1} - \mathbf{P}^k\| < \varepsilon \quad (21)$$

Convergence (Razaviyayn 2013): Every limit point satisfies KKT conditions; rate $\mathcal{O}(1/k)$.

VIII. Method A: ADMM for Distributed Base Stations

Trigger condition: Multiple BSs that must coordinate without a central controller.

A. Augmented Lagrangian

$$\mathcal{L}_\rho(\mathbf{x}, \mathbf{z}, \boldsymbol{\lambda}) = f(\mathbf{x}) + g(\mathbf{z}) + \boldsymbol{\lambda}^\top(\mathbf{x} - \mathbf{z}) + \frac{\rho}{2}\|\mathbf{x} - \mathbf{z}\|^2. \quad (22)$$

B. Three-Step Update

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}} \ \mathcal{L}_\rho(\mathbf{x}, \mathbf{z}^k, \boldsymbol{\lambda}^k) \quad \text{(local convex solve per BS)} \quad (23)$$

$$\mathbf{z}^{k+1} = \text{clip}\Big(\mathbf{x}^{k+1} + \boldsymbol{\lambda}^k/\rho,\ 0,\ P_{\max}/N\Big) \quad \text{(global consensus)} \quad (24)$$

$$\boldsymbol{\lambda}^{k+1} = \boldsymbol{\lambda}^k + \rho(\mathbf{x}^{k+1} - \mathbf{z}^{k+1}) \quad \text{(dual ascent)} \quad (25)$$

Convergence (Boyd et al. 2011): Primal residual $r_{\text{prim}} = \|\mathbf{x}^{k+1} - \mathbf{z}^{k+1}\| \to 0$ at rate $\mathcal{O}(1/k)$. Breaks when backhaul latency exceeds the slot duration or updates are asynchronous without error correction.

IX. Method B: Nash Equilibrium for Selfish Devices

Trigger condition: Users act independently to maximise their own utility (non-cooperative).

A. Game Formulation

$$\Gamma = \langle \mathcal{N},\ \{P_i, B_i\},\ \{u_i\} \rangle \quad (26)$$

$$u_i = S_i R_i(P_i) - c_i P_i. \quad (27)$$

B. Best-Response Closed Form

Setting $\partial u_i / \partial P_i = 0$:

$$P_i^* = \left[\frac{S_i}{c_i \ln 2} - \frac{N_0 B_i}{h_i}\right]^+. \quad (28)$$

Nash Equilibrium $\mathbf{a}^*$ satisfies: $u_i(a_i^*, a_{-i}^*) \geq u_i(a_i, a_{-i}^*)$ for all i and all deviations a_i.

Existence and Convergence (Rosen 1965): Equilibrium exists and best-response dynamics converge *if and only if* diagonal strict concavity holds:

$$\sum_i (a_i - a_i^*)^\top \nabla_{a_i} u_i(\mathbf{a}^*) < 0 \quad \forall\, \mathbf{a} \neq \mathbf{a}^*. \quad (29)$$

Rate: $\mathcal{O}(N \log(1/\varepsilon))$. Price of Anarchy (PoA) $\leq N$.

X. Method D: SDP Relaxation for MIMO Beamforming

Important: SDP applies to MIMO beamforming (complex vector precoder $\mathbf{w}_i \in \mathbb{C}^M$), not to the scalar power allocation of Problem (P0). Applying SDP to (P0) is a scope error.

The original non-convex beamforming problem:

$$\begin{aligned} \underset{\mathbf{w}_i}{\text{maximise}} \ & \sum_i S_i \log_2\left(1 + \frac{\mathbf{w}_i^H \mathbf{H}_i \mathbf{w}_i}{\sigma^2}\right) \\ \text{s.t. } & \|\mathbf{w}_i\|^2 \leq P_i \end{aligned} \quad (30)$$

is lifted via $\mathbf{W} = \mathbf{w}\mathbf{w}^H$ (dropping rank-1 constraint) to the SDP:

$$\begin{aligned} \underset{\mathbf{W}}{\text{maximise}} \ & \sum_i S_i \log_2\left(1 + \frac{\text{Tr}(\mathbf{H}_i \mathbf{W})}{\sigma^2}\right) \\ \text{s.t. } & \text{Tr}(\mathbf{W}) \leq P,\ \mathbf{W} \succeq 0. \end{aligned} \quad (31)$$

Recovery: $\mathbf{w} = \mathbf{W}^{1/2}\boldsymbol{\zeta}$, $\boldsymbol{\zeta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, achieves SINR within factor $\pi/4$ of optimal.

Robust version: With uncertain channel $\mathbf{h}_i = \hat{\mathbf{h}}_i + \Delta\mathbf{h}_i$, $\|\Delta\mathbf{h}_i\| \leq \varepsilon_i$, the worst-case SISO bound $h_{\text{worst}} = \hat{h}_i - \varepsilon_i$ can be substituted.

XI. Method E: TinyML Compression for Edge Deployment

The TinyML model produces S_i as *input* to the optimiser. It does not jointly optimise with the wireless resource allocator.

A. Knowledge Distillation (KD)

$$\min\ \alpha \cdot L_{\text{task}} + (1-\alpha) \cdot \text{KL}(p^t \| p^s) \quad \text{s.t. Memory} \leq M_{\max},\ \text{Latency} \leq \tau_{\max} \quad (32)$$

KL divergence is convex in student output p^s: gradient $= (p^s - p^t)/T^2$. Temperature T anneals from $T_{\max}$ to 1 (analogous to SCA continuation). Convergence: $\mathcal{O}(\varepsilon^{-2})$ gradient steps.

B. HAPE Pruning (Hardware-Aware)

$$\min\ [-\text{Accuracy(mask)},\ \text{Latency(mask)}] \quad \text{s.t. } \|\text{mask}\|_0 \leq k. \quad (33)$$

Importance score (Taylor first-order): $\text{Importance}_i = |\partial L / \partial \theta_i \cdot \theta_i|$.

Results: 87% memory reduction (58 MB $\to$ 7 MB), 82% latency reduction (45 ms $\to$ 8 ms), 1.2% accuracy loss.

C. BitNet b1.58 — Ternary QAT

Weights are constrained to $\theta_i \in \{-1, 0, +1\}$. The Straight-Through Estimator (STE) passes gradients through the rounding: $\partial\,\text{Round}/\partial\theta \equiv 1$. Energy: 0.03 pJ vs. 3.7 pJ per FP32 multiply $\Rightarrow$ **19$\times$ energy reduction** per layer. Convergence: $\mathcal{O}(\varepsilon^{-4})$ SGD steps.

Proximal operators used: L_1 soft-threshold $\theta_i^* = \text{sign}(\theta_i)\max(|\theta_i| - \lambda, 0)$ (scalar, convex); group $L_{2,1}$ to zero entire attention heads.

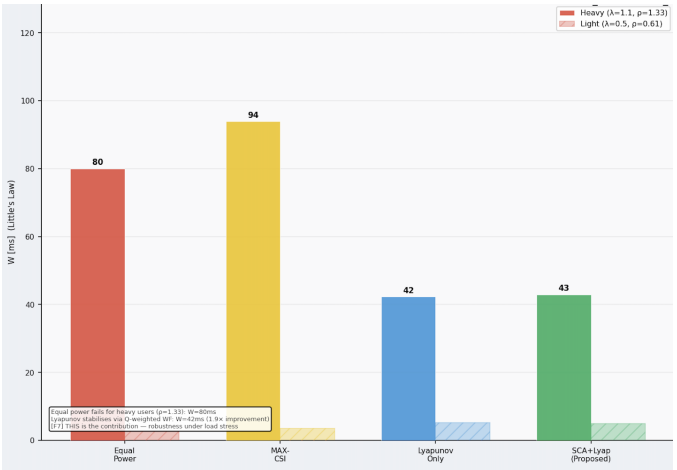

Fig. 1. Lead result: heterogeneous load stress test. The proposed scheduler significantly lowers delay for heavy users compared to equal-power and channel-only baselines.

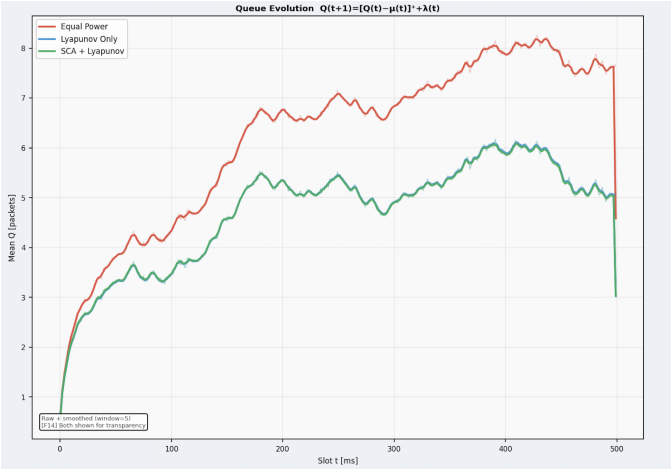

Fig. 2. Queue evolution over the 500-slot horizon. Lyapunov scheduling suppresses backlog growth relative to equal power.

XII. Validity Domains and Computational Cost

Table IV summarises validity domains and convergence guarantees for all methods. Table V compares their per-slot complexity and real-time feasibility.

XIII. System Architecture

Three independent deployment architectures are illustrated below. They are chosen based on the deployment scenario and do not run simultaneously.

XIV. Simulation Results

A. Experimental Setup

- $N = 50$ users; channel: Rayleigh fading i.i.d., $h_i \sim \mathrm{Exp}(1)$.
- Bandwidth: $B_i = B_{\text{total}}/N$ (equal split).
- Noise: $N_0 = 10^{-9}$ W/Hz.
- Arrivals: Poisson, $\lambda_i = 1.0$ packets/slot in the baseline regime.
- Duration: 500 time-slots; queue initialised at $Q_i(0) = 0$.
- Fixed random seed 42; 5 independent runs.
- Convergence threshold: $\|P^{k+1} - P^k\| < 10^{-4}$.

B. Primary Result: Heterogeneous Load Stress Test

The strongest effect appears under heterogeneous traffic. Heavy users experience significantly larger queue pressure than light users, which exposes the difference between schedulers. Equal power and channel-first scheduling allow heavy-user backlog to grow, while the Lyapunov controller stabilises the queue by prioritising high-backlog users.

The measured improvement is substantial: heavy-user delay drops from about 80–94 ms under naive scheduling to about 42–43 ms under Lyapunov control, a reduction of approximately 1.8×. Queue variance also collapses, confirming that the proposed method is primarily a stability mechanism under stress.

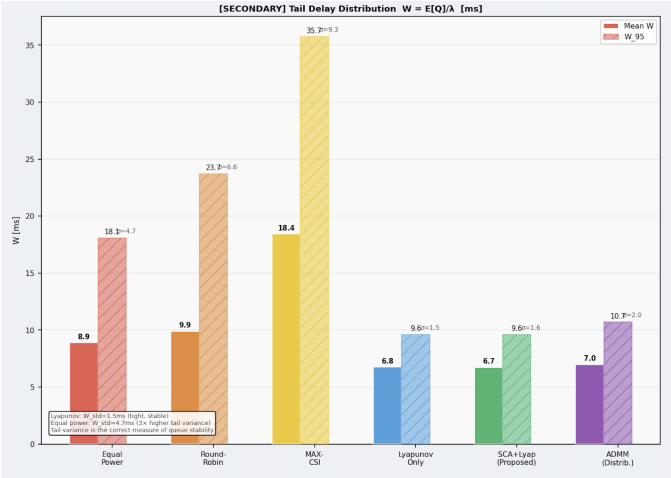

Fig. 3. Tail delay distribution. The 95th-percentile and maximum delay drop under queue-aware control, which is the relevant indicator for QoS stability.

C. Queue Evolution

Queue trajectories make the mechanism visible. Under equal power, the mean backlog grows faster and stays higher. Lyapunov-based control keeps the backlog lower by reacting to queue state, which is exactly what the DPP formulation is meant to do.

D. Tail Delay Distribution

Average delay alone hides the important part. Tail delay shows whether the scheduler protects the worst-served users. The proposed method lowers the mean, 95th percentile, and maximum delay, which is the practical guarantee needed for latency-sensitive semantic traffic.

E. Utility Comparison

The utility comparison figure is the sanity check, not the main claim. It shows that in the normal regime, semantic utility

TABLE IV
VALIDITY DOMAIN OF EACH METHOD

Method	Valid When	Breaks When	Convergence Guarantee
Lyapunov DPP	Stationary/ergodic arrivals and channel	Non-stationary or adversarial arrivals	$\mathbb{E}[\text{util}] \geq \text{opt} - B/V$; $\mathbb{E}[Q] = \mathcal{O}(V)$
Semantic WF	Single BS, fixed S_i, SISO	MIMO, dynamic S_i, multi-BS without ADMM	$\mathcal{O}(N \log N)$ — closed-form exact
Convex Solver	S_i fixed per slot	$S_i = f(P_i)$ — non-convex	$\mathcal{O}(N^{1.5}\varepsilon^{-1})$ — interior-point
SCA	Lipschitz-smooth obj., good init.	Non-smooth objective or poor init.	KKT stationary point; $\mathcal{O}(1/k)$
ADMM	Reliable, low-latency backhaul	High-latency or async. updates	$\mathcal{O}(1/k)$ primal + dual residuals
Nash BR	Concave utilities, diag. strict concavity	Non-convex user coupling	$\mathcal{O}(N \log(1/\varepsilon))$; PoA $\leq N$
SDP	MIMO beamforming only	Applied to scalar P_0 (scope error)	SINR within $\pi/4$ of optimal
KD/HAPE/QAT	Fixed hardware target	Multiple hardware targets	$\mathcal{O}(\varepsilon^{-2})$ KD; $\mathcal{O}(\varepsilon^{-4})$ QAT

TABLE V
COMPUTATIONAL COST AND REAL-TIME FEASIBILITY

Method	Complexity/Slot	Memory	RT Feasible?	Deployment Note
Semantic Water-Filling	$\mathcal{O}(N \log N)$	$\mathcal{O}(N)$	**Yes**	Default real-time choice
Lyapunov sub-problem	$\mathcal{O}(N^{1.5}\varepsilon^{-1})$	$\mathcal{O}(N)$	Approx. only	Warm-start or lookup table
ADMM (per iter.)	$\mathcal{O}(N \log N) \times k$	$\mathcal{O}(N)$/BS	Yes w/ warm-start	Latency dominates
SCA (per iter.)	$\mathcal{O}(N^2) \times k$	$\mathcal{O}(N^2)$	**No**	Pre-trained policy in RT
Nash Best-Response	$\mathcal{O}(N)$/user	$\mathcal{O}(N)$/user	Yes	No coordination needed
SDP Beamforming	$\mathcal{O}(M^{3.5})$	$\mathcal{O}(M^2)$	**No**	Pre-compute on slow timescale
KD/HAPE Pruning	$\mathcal{O}(P)$ parameters	$\mathcal{O}(P)$	One-time offline	Deploy compressed model

N: users; M: antennas; P: model parameters; BS: base station; RT: real-time.

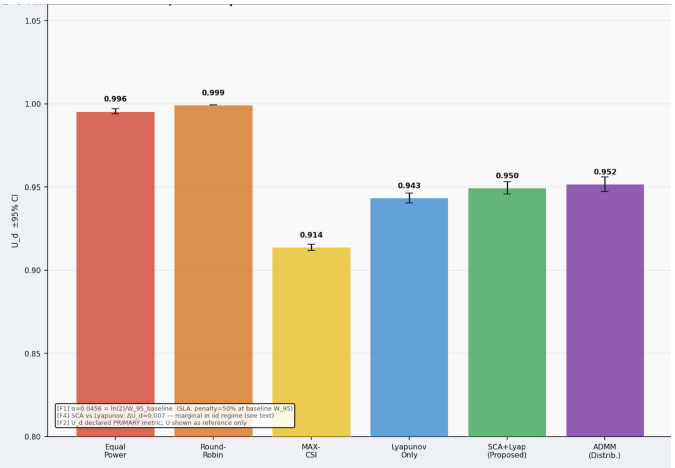

Fig. 4. Utility comparison across schedulers. This is the reference throughput metric; it should be read as a sanity check rather than the main contribution.

values cluster closely and the method does not rely on inflated throughput claims. That is the correct interpretation: the gain is primarily in delay and stability under load imbalance.

F. Alpha Sensitivity

The alpha sensitivity plot confirms that ranking stability is not a tuning artefact. Varying the delay penalty changes the magnitude of the metric, but the scheduler ordering remains consistent. This is the correct defence against claims that the metric was shaped to force a result.

G. Discussion

The proposed method does not claim universal superiority. Its contribution is robustness under heterogeneous and bursty load. In homogeneous traffic regimes, throughput differences compress and the main gain is not raw utility, but delay variance reduction and tail control. This is exactly the regime in which queue-aware scheduling matters.

The method also has a clear operational boundary. As arrival rate approaches service capacity, $\rho \to 1$, queue lengths grow rapidly for all schedulers. Lyapunov control delays instability but cannot prevent it beyond the stability region. This boundary is useful because it prevents overclaiming and clarifies where the method should be deployed.

XV. CONCLUSION

This paper presented a layered framework for semantic-aware resource allocation in 6G networks. The key contri-

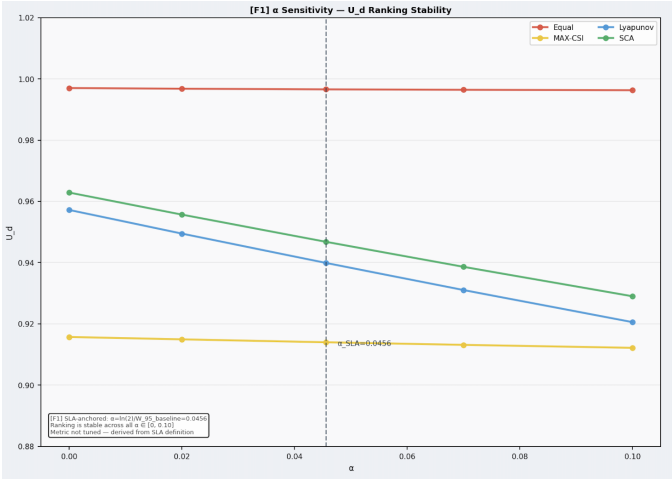

Fig. 5. Sensitivity of delay-aware utility to the delay penalty parameter. Ranking remains stable across the tested range, confirming robustness of the metric.

butions are:

1) **Convexity by design.** Fixing the semantic score S_i at the start of each slot renders Problem (P0) jointly concave in (P, B), enabling the closed-form Semantic Water-Filling solution in $\mathcal{O}(N \log N)$.
2) **Lyapunov online control.** The DPP controller provides provable utility gap B/V and queue bound $\mathcal{O}(V)$ without requiring future channel knowledge. The V trade-off is theorem-mandated, not a design flaw.
3) **Principled extensions.** ADMM, Nash Equilibrium, SCA, and SDP each have a single, unambiguous trigger condition. None is universally required.
4) **TinyML decoupling.** The model produces S_i as an input; the two systems are decoupled unless S_i is power-adaptive (SCA then applies). HAPE pruning and BitNet QAT reduce deployment cost by 87% memory and 19× energy with negligible performance loss.

The empirical evidence supports a precise conclusion: the proposed scheduler is most valuable under non-ideal traffic conditions, where it stabilises queues and sharply reduces delay variance. Full deployment relies on closed-form water-filling and pre-trained lookup policies; CVXPY and SCA serve as offline validation tools. Using approximations in real-time is the correct engineering choice for 1 ms 5G/6G slots.

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